

## Create dictionary

```
In [121]: mydict = {}
```

```
In [122]: mydict
```

```
Out[122]: {}
```

```
In [123]: type(mydict)
```

```
Out[123]: dict
```

```
In [4]: mydict = {1:'one', 2:'two', 3:'three'} # dictionary with int keys
```

```
In [5]: mydict
```

```
Out[5]: {1: 'one', 2: 'two', 3: 'three'}
```

```
In [6]: mydict = dict({1: 'one', 2: 'two', 3: 'three'})
```

```
In [7]: mydict
```

```
Out[7]: {1: 'one', 2: 'two', 3: 'three'}
```

```
In [11]: mydict2 = {'1': 'one', 'B1': 'two', 'C1': 'three'}
```

```
In [12]: mydict2
```

```
Out[12]: {'1': 'one', 'B1': 'two', 'C1': 'three'}
```

```
In [13]: mydict2.keys()
```

```
Out[13]: dict_keys(['1', 'B1', 'C1'])
```

```
In [14]: mydict2.values()
```

```
Out[14]: dict_values(['one', 'two', 'three'])
```

```
In [15]: mydict2.items()
```

```
Out[15]: dict_items([('1', 'one'), ('B1', 'two'), ('C1', 'three')])
```

```
In [17]: mydict2
```

```
Out[17]: {'1': 'one', 'B1': 'two', 'C1': 'three'}
```

```
In [19]: #accessing items
```

```
In [20]: mydict = {1:'one', 2:'two', 3:'three', 4:'four', 6:'six'}
```

```
In [21]: mydict
```

```
Out[21]: {1: 'one', 2: 'two', 3: 'three', 4: 'four', 6: 'six'}
```

```
In [22]: mydict[1]
```

```
Out[22]: 'one'
```

```
In [23]: mydict[6]
```

```
Out[23]: 'six'
```

```
In [25]: mydict.get(1) #accessing item using get() method
```

```
Out[25]: 'one'
```

```
In [27]: mydict3 = {'Name': 'Ahmed', 'ID' : 45877, 'Designation': 'Analyst', 'Salary': 45
```

```
In [28]: mydict3
```

```
Out[28]: {'Name': 'Ahmed', 'ID': 45877, 'Designation': 'Analyst', 'Salary': 450000}
```

```
In [30]: mydict3.get('Salary') #using get() method
```

```
Out[30]: 450000
```

```
In [33]: mydict3['ID'] #accessing value using key
```

```
Out[33]: 45877
```

```
In [35]: mydict3['Name']
```

```
Out[35]: 'Ahmed'
```

### Add, Remove & Change Items

```
In [53]: mydict4 = {'Name': 'Ahmed', 'ID': 45877, 'Designation': 'Analyst', 'Salary': 450
```

```
In [54]: mydict4
```

```
Out[54]: {'Name': 'Ahmed',  
          'ID': 45877,  
          'Designation': 'Analyst',  
          'Salary': 450000,  
          'Address': 'New York'}
```

```
In [55]: mydict4['ID'] = 33787 #Changing Dictionary Items  
mydict4['Address'] = 'Andaman'
```

```
In [56]: mydict4
```

```
Out[56]: {'Name': 'Ahmed',  
          'ID': 33787,  
          'Designation': 'Analyst',  
          'Salary': 450000,  
          'Address': 'Andaman'}
```

```
In [57]: mydict4 = {'ID':19995}  
mydict4.update(mydict4) #Changing dictionary items using update() function
```

```
In [58]: mydict4
```

```
Out[58]: {'ID': 19995}
```

```
In [59]: mydict4['Age'] = 25
```

```
In [60]: mydict4
```

```
Out[60]: {'ID': 19995, 'Age': 25}
```

```
In [61]: mydict4
```

```
Out[61]: {'ID': 19995, 'Age': 25}
```

```
In [62]: mydict4 = {'Name': 'Ahmed', 'ID': 45877, 'Designation': 'Analyst', 'Salary': 450000, 'Address': 'New York'}
```

```
In [63]: mydict4
```

```
Out[63]: {'Name': 'Ahmed',  
          'ID': 45877,  
          'Designation': 'Analyst',  
          'Salary': 450000,  
          'Address': 'New York'}
```

```
In [64]: mydict4['Age'] = 25 #adding an item into the dictionary
```

```
In [69]: mydict4
```

```
Out[69]: {'Name': 'Ahmed',  
          'Designation': 'Analyst',  
          'Salary': 450000,  
          'Address': 'New York',  
          'Age': 25}
```

```
In [70]: mydict4.popitem() # a random item is removed
```

```
Out[70]: ('Age', 25)
```

```
In [71]: mydict4
```

```
Out[71]: {'Name': 'Ahmed',  
          'Designation': 'Analyst',  
          'Salary': 450000,  
          'Address': 'New York'}
```

```
In [72]: del[mydict4['Salary']]
```

```
In [73]: mydict4
```

```
Out[73]: {'Name': 'Ahmed', 'Designation': 'Analyst', 'Address': 'New York'}
```

```
In [74]: mydict4.clear()
```

```
In [75]: mydict4
```

```
Out[75]: {}
```

```
In [78]: del mydict4 #Delete the dictionary object
```

```
-----  
NameError                                Traceback (most recent call last)  
Cell In[78], line 1  
----> 1 del mydict4 #Delete the dictionary object  
  
NameError: name 'mydict4' is not defined
```

```
In [79]: mydict4
```

```
-----  
NameError                                Traceback (most recent call last)  
Cell In[79], line 1  
----> 1 mydict4  
  
NameError: name 'mydict4' is not defined
```

Copy Dictionary

```
In [80]: mydict = {'Name': 'Anand' , 'ID': 12345 , 'DOB': 1981 , 'Address' : 'Hilsinki'}  
mydict
```

```
Out[80]: {'Name': 'Anand', 'ID': 12345, 'DOB': 1981, 'Address': 'Hilsinki'}
```

```
In [82]: mydict1 = mydict
```

```
In [83]: id(mydict), id(mydict1)
```

```
Out[83]: (1330382640640, 1330382640640)
```

```
In [84]: mydict2 = mydict.copy()
```

```
In [85]: mydict2
```

```
Out[85]: {'Name': 'Anand', 'ID': 12345, 'DOB': 1981, 'Address': 'Hilsinki'}
```

```
In [86]: mydict1
```

```
Out[86]: {'Name': 'Anand', 'ID': 12345, 'DOB': 1981, 'Address': 'Hilsinki'}
```

```
In [87]: mydict3 = mydict2
```

```
In [88]: mydict3
```

```
Out[88]: {'Name': 'Anand', 'ID': 12345, 'DOB': 1981, 'Address': 'Hilsinki'}
```

```
In [89]: id(mydict2)
```

```
Out[89]: 1330382605760
```

Loop through a dictionary

```
In [92]: mydict1 = {'Name': 'Asif' , 'ID': 12345 , 'DOB': 1991 , 'Address' : 'Hilsinki'}
```

```
In [93]: mydict1
```

```
Out[93]: {'Name': 'Asif', 'ID': 12345, 'DOB': 1991, 'Address': 'Hilsinki'}
```

```
In [97]: for i in mydict1:  
         print(i, ': ', mydict1[i])
```

```
Name : Asif  
ID : 12345  
DOB : 1991  
Address : Hilsinki
```

```
In [100]: for i in mydict1:  
          print(mydict1[i])
```

```
Asif  
12345  
1991  
Hilsinki
```

Dictionary Membership

```
In [101]: mydict1
```

```
Out[101]: {'Name': 'Asif', 'ID': 12345, 'DOB': 1991, 'Address': 'Hilsinki'}
```

```
In [102]: 'ID' in mydict1
```

```
Out[102]: True
```

```
In [103]: 'Name' in mydict1
```

```
Out[103]: True
```

All / Any

```
In [104]: mydict1
```

```
Out[104]: {'Name': 'Asif', 'ID': 12345, 'DOB': 1991, 'Address': 'Hilsinki'}
```

```
In [105]: all(mydict1)
```

```
Out[105]: True
```

```
In [106]: any(mydict1)
```

```
Out[106]: True
```

```
In [ ]:
```